CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

KRASNOKAMENSK URANIUM HUB

Russia-China Energy — Mining — Uranium Processing

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CONVERSION TIER

CONSTRAINED

NODE ID	27 — SENI-04
LOCATION	Zabaykalsky Krai, Russia
COORDINATES	50.10°N, 118.03°E
OPERATOR	ARMZ Uranium Holding / Rosatom (state)
NETWORK GROUP	Russia-China Energy
PROCESS ROLE	Upstream — Uranium Mining and Milling (U_3O_8)
STATUS	CONSTRAINED — EO 13382 authority available; nuclear safety argument constrains near-term action

KEY JUDGMENTS

KJ 1

PIMCU produces approximately 90% of Russia's domestic uranium ore. ARMZ Uranium Holding is legally designatable under EO 13382 as a Rosatom subsidiary [TIER 1 — Rosatom 2023 Annual Report]. The principal constraint is TVEL's active fuel supply contracts for two operating Chinese civilian reactors.

LIKELY — HIGH CONFIDENCE

KJ 2

TVEL JSC's Tianwan and Xudapu NPP fuel supply contracts create a nuclear safety argument under IAEA Article III(A) that Beijing would exploit in multilateral forums if ARMZ is designated without DOE/NNSA coordination.

VERY LIKELY — MODERATE-HIGH CONFIDENCE

KJ 3	Post-2022 migration of TVEL-CNNC contracts to CNY-denominated settlement has materially reduced U.S. financial enforcement leverage at this node. The public record does not permit a firm conclusion on the current extent of residual USD-denominated ARMZ banking activity. **LIKELY — MODERATE CONFIDENCE**
KJ 4	The enforcement ceiling at Krasnokamensk is financial disruption of ARMZ's residual Western banking access rather than full SDN designation. A threshold condition — defined by Tianwan/Xudapu fuel stockpile lead time — has not been established in public reporting. **LIKELY — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED**

I. FACILITY OVERVIEW

ATTRIBUTION	Rosatom annual reports and IAEA safeguards documentation identify PIMCU at Krasnokamensk as Russia's largest uranium mining complex, producing approximately 90% of domestic uranium ore. ARMZ Uranium Holding — a wholly-owned Rosatom subsidiary — operates the facility [TIER 1 — Rosatom 2023 Annual Report; IAEA Country Nuclear Power Profiles, Russia 2024].
JUDGMENT	Official reporting supports a high-confidence judgment that PIMCU is the primary feedstock origin node for the Russian nuclear fuel cycle. The facility's geographic proximity to the Russia-China border (~70 km) and its role supplying Tianwan and Xudapu NPPs make it structurally significant in the Russia-China energy relationship.
LIMITS	The public record does not establish current annual production tonnage with precision. Published Tier 2 figures range from 1,800 to 2,500 tonnes uranium ore concentrate annually. The public record does not permit a firm conclusion on specific contract volumes for Chinese reactor fuel supply.

II. FUEL CYCLE CONNECTIONS

CONNECTION	DETAIL
Yellowcake → Conversion	U_3O_8 → UF_6 conversion at Seversk (SCC, Node 29) or Angarsk (AECC, Node 31)
Conversion → Enrichment	UF_6 → centrifuge enrichment at Novouralsk (UECC, Node 28), Zelenogorsk (ECP, Node 30), Seversk, or Angarsk
Enrichment → Fabrication	LEU → fuel assembly fabrication at Elektrostal (Node 32) or Novosibirsk NCCP (Node 33)

Declared civilian end-use	Tianwan NPP and Xudapu NPP (China) via TVEL JSC-CNNC fuel supply contracts

III. FINANCIAL NETWORK

ATTRIBUTION	Rosatom consolidated financial reporting indicates ARMZ Uranium Holding routes production revenue through TVEL JSC for downstream processing and international fuel sales. OFAC designated TVEL JSC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List, April 2024]. ARMZ Uranium Holding remains undesignated as of March 2026.
JUDGMENT	Available reporting supports a moderate-confidence judgment that post-2022 TVEL-CNNC contract settlement has migrated substantially to CNY-denominated clearing through Chinese state banks, materially reducing Western financial leverage at this node.
LIMITS	The public record does not establish specific CNY clearing architecture for TVEL-CNNC transactions, the current status of any residual USD-denominated ARMZ banking relationships, or the contract value of the Xudapu fuel supply agreement.

IMPLEMENTATION FRICTION Post-2022 CNY settlement migration substantially insulates TVEL-CNNC transactions from U.S. secondary sanctions without targeting Chinese clearing banks (ICBC, Bank of China) directly — a major escalation not yet executed.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	ARMZ designatable as Rosatom subsidiary. Legal predicate available.
CAATSA §231	Defense/Intel Sector	Rosatom state nuclear corporation; sector authority applies to ARMZ.
IEEPA	Broad Economic Auth.	Available for emergency expansion if fuel diversion evidence emerges.
DOE/NNSA	Policy Coordination	Interagency consultation required given active Chinese reactor fuel supply.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC designated TVEL JSC — ARMZ's commercial parent — under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. ARMZ Uranium Holding remains undesignated. EO 13382 identifies WMD proliferators and their support networks as designation targets.
JUDGMENT	Official reporting supports a moderate-high-confidence judgment that ARMZ is legally designatable under EO 13382 as a Rosatom subsidiary. The enforcement constraint is operational consequence, not legal authority: designating the fuel supply chain for two operating Chinese civilian reactors creates a nuclear safety argument under IAEA Article III(A) that Beijing would exploit in multilateral forums.
LIMITS	The public record does not establish whether the nuclear safety constraint is permanent or threshold-conditional. No publicly available analysis quantifies the fuel stockpile lead time at Tianwan or Xudapu that would define an actionable designation window.

DOCUMENTED STRUCTURAL CONSTRAINT IAEA Article III(A) and the multilateral safeguards framework create a documented structural constraint on ARMZ designation that operates independent of U.S. domestic legal authority. DOE/NNSA and allied nuclear regulator engagement is the precondition.

VI. ENFORCEMENT GAPS

Gap 1 — No Secondary Enforcement Instrument Identified

The enforcement theory identifies TVEL designation as constrained but does not specify a viable secondary vector. Shipping intermediaries and financial institutions clearing TVEL-CNNC contracts are candidate secondary targets; neither has been mapped or designated.

Gap 2 — Threshold Condition for Action Undefined

The public record does not establish the threshold condition under which the nuclear safety constraint lifts. ANALYTICALLY INFERRED with LOW-MODERATE CONFIDENCE that a Tianwan/Xudapu fuel stockpile lead time of 12–18 months would constitute a defensible action window. This inference is not supported by Tier 1 or Tier 2 sources.

Gap 3 — ARMZ Residual Western Banking Access Unmapped

Specific Western banking relationships maintained by ARMZ Uranium Holding post-2022 are not documented in open-source analysis. This is the primary financial intelligence gap at this node.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	Residual ARMZ Western banking channel unmapped. CNY

		migration confirmed; specific clearing banks for residual USD exposure not identified in OSINT. Upgrade: TENEX Geneva-entity Swiss Handelsregister search.
2	THRESHOLD	Fuel stockpile lead time for Chinese reactor operators dependent on ARMZ supply not established. Upgrade: Classified energy contract data or IAEA fuel cycle monitoring.

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NOVOURALSK — URAL ELECTROCHEMICAL COMBINE

Enrichment — Closed City (Sverdlovsk-44)

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CONVERSION TIER

TARGETABLE

NODE ID	28 — SENI-05
LOCATION	Novouralsk, Sverdlovsk Oblast, Russia (ZATO)
COORDINATES	57.26°N, 60.10°E
OPERATOR	UECC / TVEL JSC / Rosatom
NETWORK GROUP	Russian Nuclear Fuel Cycle — Enrichment Tier
PROCESS ROLE	Mid-Cycle — Uranium Enrichment (Gas Centrifuge, LEU; historically HEU)
STATUS	TARGETABLE — designation legally available; allied supply precondition not yet met

KEY JUDGMENTS

KJ 1

UECC accounts for approximately 48% of Russian centrifuge enrichment capacity and more than 20% of global enrichment supply [TIER 1 — Rosatom 2023 Annual Report; WNA Enrichment Overview 2024]. OFAC designated TVEL JSC — UECC's commercial parent — on April 19, 2024. UECC is not explicitly listed on the SDN list as of March 2026.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Full UECC SDN designation without allied coordination and supply substitution would create acute LEU shortages for reactor operators in Eastern Europe, Asia, and the U.S. market. Urenco, Orano, and Centrus combined capacity does not meet the

	≥85% UECC displacement threshold as of March 2026. **LIKELY — MODERATE-HIGH CONFIDENCE**
KJ 3	TVEL's April 2024 SDN designation constructively covers UECC under the 50% ownership rule. The absence of an explicit SDN listing creates compliance ambiguity that sophisticated actors exploit to continue transacting through non-designated intermediaries. **LIKELY — MODERATE CONFIDENCE**
KJ 4	Secondary financial pressure on TVEL's correspondent banking — targeting SWU revenue clearing infrastructure — is actionable now without triggering supply disruption and should precede direct UECC designation. **LIKELY — MODERATE CONFIDENCE**
KJ 5	UECC historically produced HEU for Soviet weapons programs using the same centrifuge cascade infrastructure. The public record does not establish current HEU production at UECC; LEU civilian production is the assessed current operational posture. **UNLIKELY (CURRENT HEU) — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED**

I. FACILITY OVERVIEW

ATTRIBUTION	Rosatom's 2023 Annual Report identifies UECC as the largest single enrichment facility in Russia. The World Nuclear Association's Uranium Enrichment Overview (2024) reports Russia provides approximately 44% of global uranium enrichment services [TIER 1 — Rosatom 2023 Annual Report; WNA 2024]. The facility is located in Novouralsk — formerly Sverdlovsk-44 — a ZATO closed city with access restricted under Russian federal law.
JUDGMENT	Official reporting supports a high-confidence judgment that UECC is the single most strategically significant node in the Russian fuel cycle from a global market perspective. Enforcement action targeting UECC would have immediate consequences for enrichment supply to reactor operators in more than 12 countries.
LIMITS	The public record does not establish the current centrifuge generation operating at UECC, whether HEU production capacity has been retained or eliminated post-Megatons-to-Megawatts (completed 2013), or the specific cascade configuration allocable to civilian versus potential weapons-grade production.

II. FUEL CYCLE CONNECTIONS

CONNECTION	DETAIL

UF_6 feedstock input	Received from Seversk (SCC, Node 29) or Angarsk (AECC, Node 31) conversion facilities
LEU → Domestic fabrication	Enriched output → Elektrostal (Node 32, VVER/RBMK) and Novosibirsk NCCP (Node 33)
LEU → EU export (disrupted)	TVEL contracts with Eastern European VVER operators; partially disrupted post-SDN designation
LEU → China (TVEL-CNNC)	Active contracts; CNY-denominated settlement post-2022
LEU → India/other	NPCIL and other Asian operators; INR-RUB settlement post-2022
Historical HEU	Megatons-to-Megawatts program completed 2013; Rosatom manages residual HEU inventory

III. FINANCIAL NETWORK

ATTRIBUTION	TVEL JSC holds all commercial enrichment service contracts for UECC SWU output and was designated by OFAC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. The Congressional Research Service's 2023 analysis reported TVEL's annual enrichment revenue was estimated at $2–3 billion prior to sanctions [TIER 1 — CRS R47742, 2023].
JUDGMENT	Available reporting supports a moderate-confidence judgment that post-2022 and post-SDN-designation, TVEL has migrated significant SWU contract volume to CNY and bilateral currency clearing. Residual Western banking activity may continue through OFAC General License carve-out wind-down provisions. The public record does not permit a firm conclusion on the current clearing status of all active TVEL contracts.
LIMITS	The public record does not establish which specific correspondent banks are currently clearing residual TVEL SWU revenue, the terms of active OFAC General License carve-outs for UECC-derived enrichment services, or the total CNY-denominated SWU revenue volume post-2022.

IMPLEMENTATION FRICTION Post-2022 CNY migration substantially reduces leverage available through Western correspondent banking restrictions for Chinese-destined SWU revenue. The financial enforcement node is TVEL JSC — not UECC directly. TVEL's SPV network in Geneva, Dubai, and Istanbul is the primary residual enforcement target.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	TVEL SDN April 2024. UECC designatable as TVEL/Rosatom subsidiary under 50% rule or explicit listing.
CAATSA §231	Defense/Intel Sector	Rosatom nuclear sector; UECC directly within scope.
IEEPA / OFAC	Entity Designation	UECC not explicitly listed. Independent designation action available.
NSG	Export Control Coordination	Allied coordination framework available for enrichment technology controls.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC designated TVEL JSC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. UECC is constructively blocked as a TVEL subsidiary under the 50% ownership rule but is not explicitly listed on the SDN list as of March 2026.
JUDGMENT	Official reporting supports a moderate-high-confidence judgment that UECC is legally targetable and UECC designation is operationally significant. The binding constraint is market consequence: full designation without allied coordination and supply substitution creates acute LEU shortages for NATO-allied reactor operators in Eastern Europe. This is a sequencing constraint, not a legal barrier.
LIMITS	The public record does not establish the current timeline for Urenco, Orano, and Centrus capacity to reach the ≥85% UECC displacement threshold. Publicly available projections are Tier 2 at best; the supply precondition cannot be confirmed as met or unmet with HIGH CONFIDENCE.

DOCUMENTED STRUCTURAL CONSTRAINT Allied enrichment supply capacity does not meet the precondition for full UECC designation without unacceptable reactor safety risk to Eastern European NATO allies. This is a documented structural — not legal — constraint. Enforcement sequencing should account for the supply buildout timeline.

IMPLEMENTATION FRICTION TVEL's April 2024 SDN designation has not been fully enforced against legacy contracts. Some EU and Asian operators appear to be continuing enrichment service procurement under carve-out arrangements or third-country intermediaries. Carve-out compliance audit is a prerequisite for any escalated action.

VI. ENFORCEMENT GAPS

Gap 1 — Allied Supply Capacity Precondition Unmet

Urenco, Orano, and Centrus combined cannot fully replace UECC output as of March 2026 [MODERATE CONFIDENCE — ANALYTICALLY INFERRED from publicly available capacity announcements; Tier 2 sources]. Direct designation without supply substitution creates reactor safety risk for Eastern European NATO allies.

Gap 2 — Explicit SDN Listing Absent

UECC is not explicitly named on the SDN list as of March 2026. Constructive coverage under the 50% rule creates compliance ambiguity that sophisticated third-country financial intermediaries exploit to continue transacting without clear OFAC blocking obligation.

Gap 3 — Legacy Contract Carve-Out Compliance Unaudited

The public record does not establish whether all OFAC General License wind-down periods for TVEL legacy contracts have expired or whether ongoing UECC-derived SWU clearing through Western banks continues. This is a critical enforcement gap.

Gap 4 — Post-Designation Revenue Monitoring

Post-2022 migration to non-USD clearing has reduced U.S. visibility into TVEL/UECC revenue flows. CNY-denominated SWU contracts with Chinese utilities are not accessible through Western financial monitoring infrastructure. LOW CONFIDENCE — ANALYTICALLY INFERRED that all Western-clearing UECC revenue has been eliminated.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	SPV network asserted but not evidenced in open sources. Named entities in Geneva/Dubai/Istanbul not identified. Upgrade: Swiss Handelsregister TVEL/TENEX-affiliated entity search; UAE commercial registry.
2	THRESHOLD	85% displacement precondition defined at system level (SYS-01). Monitor Urenco/Orano/Centrus expansion announcements; current trajectory meets threshold approximately 2028–2030.
3	DESIGNATION	Explicit UECC SDN listing recommended before enforcement action. Constructive TVEL coverage applies; explicit listing eliminates third-country compliance ambiguity.

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SEVERSK — SIBERIAN CHEMICAL COMBINE

Conversion / Enrichment — Closed City (Tomsk-7) — Former Weapons Pu Production

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CONVERSION TIER

TARGETABLE

NODE ID	29 — SENI-06
LOCATION	Seversk, Tomsk Oblast, Russia (ZATO)
COORDINATES	56.59°N, 84.90°E
OPERATOR	Siberian Chemical Combine (SCC) / TVEL JSC / Rosatom
NETWORK GROUP	Russian Nuclear Fuel Cycle — Conversion and Enrichment Tier
PROCESS ROLE	Mid-Cycle — UF_6 Conversion and Uranium Enrichment; Former Weapons-Grade Pu Production
STATUS	TARGETABLE — EO 13382 available on dual basis; IAEA safeguards consultation required before action

KEY JUDGMENTS

KJ 1

SCC performs UF_6 conversion and uranium enrichment at a single site within a former weapons-grade plutonium production complex. The weapons production history provides an independent EO 13382 designation predicate beyond the Rosatom subsidiary chain argument.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

SCC's ZATO status and former weapons infrastructure create potential gaps in IAEA safeguards inspection coverage. The public record does not permit a firm conclusion on whether declared civilian activities fully account for all current facility capabilities.

KJ 3	ROUGHLY EVEN ODDS — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED SCC's combined conversion and enrichment capability at a single site with former weapons infrastructure makes it the most analytically sensitive node in the Russian fuel cycle for nonproliferation assessment. LIKELY — MODERATE CONFIDENCE

I. FACILITY OVERVIEW

ATTRIBUTION	The Nuclear Threat Initiative's Russia Nuclear Profile and IAEA Country Nuclear Power Profiles identify the Siberian Chemical Combine at Seversk — formerly Tomsk-7 — as a ZATO facility conducting UF_6 conversion and uranium enrichment [TIER 2 — NTI Russia Profile 2023; TIER 1 — IAEA CNPP Russia 2024]. The facility's weapons-grade plutonium production reactors were shut down by 2008 under U.S.-Russia cooperative threat reduction agreements [TIER 1 — DOE/NNSA CTR Program reports].
JUDGMENT	Official and Tier 2 reporting supports a moderate-confidence judgment that SCC is operationally significant in both the conversion and enrichment tiers of the Russian fuel cycle, and that its dual-use infrastructure history distinguishes it analytically from purely civilian enrichment sites.
LIMITS	The public record does not establish the current operational status of former weapons infrastructure at SCC, the scope of IAEA safeguards inspection coverage at ZATO facilities with former weapons programs, or whether any capability beyond declared civilian operations is maintained.

II. FUEL CYCLE CONNECTIONS

CONNECTION	DETAIL
U_3O_8 feedstock input	Receives yellowcake from Krasnokamensk (PIMCU, Node 27) and other ARMZ mining operations
UF_6 conversion output	Internal enrichment cascades or transfer to Novouralsk (UECC, Node 28) / Zelenogorsk (ECP, Node 30)
LEU enrichment output	Fuel fabrication at Elektrostal (Node 32) and Novosibirsk NCCP (Node 33)
Historical Pu production	Weapons-grade Pu production ceased ~2008; legacy HEU inventory under Rosatom management

III. FINANCIAL NETWORK

ATTRIBUTION	SCC's civilian conversion and enrichment operations route financial flows through TVEL JSC, which was designated by OFAC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. SCC does not appear as an independent commercial entity in Western financial monitoring databases; its commercial operations are consolidated under TVEL.
JUDGMENT	Available reporting supports a low-moderate-confidence judgment that SCC has no independent Western banking relationships accessible to OFAC financial enforcement. The TVEL SDN designation constructively covers SCC's civilian revenue flows under the 50% ownership rule.
LIMITS	The public record does not establish whether SCC's former weapons infrastructure operations receive separate defense budget funding outside TVEL's commercial structure, or whether any SCC-specific financial flows are visible in TVEL consolidated disclosures.

IMPLEMENTATION FRICTION SCC's financial consolidation under TVEL means SCC-specific financial pressure cannot be applied without also affecting the broader TVEL structure. No independent financial enforcement lever exists at this node beyond TVEL designation.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators — Dual Basis	(1) TVEL/Rosatom subsidiary chain; (2) independent weapons production history predicate. Second basis is analytically stronger.
CAATSA §231	Defense/Intel Sector	SCC's defense program history places it directly within scope.
IEEPA	Broad Economic Auth.	Available if undeclared activity evidence emerges.
DOE/NNSA	CTR Legacy Knowledge	Former CTR program involvement creates residual institutional knowledge relevant to any designation action.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC designated TVEL JSC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. SCC is constructively covered as a TVEL subsidiary. DOE/NNSA's Cooperative Threat Reduction program provided monitoring access to SCC weapons infrastructure conversion through approximately 2014; that access was subsequently curtailed.
JUDGMENT	Official reporting supports a moderate-confidence judgment that SCC is designatable under EO 13382 on two independent bases: Rosatom subsidiary chain and weapons production history. The weapons history predicate is stronger for enforcement purposes because it does not require proving current weapons-related activity. The enforcement gap specific to SCC is IAEA safeguards coverage at ZATO facilities with former weapons programs.
LIMITS	The public record does not establish the current scope of IAEA inspection coverage at SCC, the disposition of HEU inventory produced prior to weapons reactor shutdown, or whether CTR-era facility characterization remains accurate as of 2026.

DOCUMENTED STRUCTURAL CONSTRAINT IAEA safeguards coverage at ZATO facilities with former weapons infrastructure has acknowledged gaps that the public record does not resolve. Designation action should be preceded by enhanced IAEA safeguards consultation to avoid designating activities that IAEA has verified as civilian.

VI. ENFORCEMENT GAPS

Gap 1 — IAEA Safeguards Coverage Gap

ZATO status and former weapons infrastructure create potential blind spots in IAEA inspection coverage at SCC. Undeclared activities would not be visible through current monitoring mechanisms. This is the highest-priority intelligence gap at this node. LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED based on known IAEA inspection limitations at closed nuclear cities.

Gap 2 — CTR Program Termination

The Cooperative Threat Reduction program provided the primary monitoring mechanism for SCC weapons infrastructure conversion. Following curtailment of CTR access, no equivalent transparency mechanism exists for former weapons facilities at this site.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	SCC has no independent Western financial profile in OSINT. TVEL consolidation obscures SCC-specific revenue. Upgrade: TVEL

		annual report subsidiary disclosures; Russian Federal Tax Service filings.
2	DESIGNATION	Explicit SCC SDN listing recommended. IAEA consultation required before designation given ZATO status and former weapons-grade infrastructure.

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ZELENOGORSK — ELECTROCHEMICAL PLANT

Enrichment / Isotope Production — Closed City (Krasnoyarsk-45)

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

CONVERSION TIER

TARGETABLE

NODE ID	30 — SENI-07
LOCATION	Zelenogorsk, Krasnoyarsk Krai, Russia (ZATO)
COORDINATES	56.11°N, 94.51°E
OPERATOR	Electrochemical Plant (ECP) / TVEL JSC / Rosatom
NETWORK GROUP	Russian Nuclear Fuel Cycle — Enrichment Tier
PROCESS ROLE	Mid-Cycle — Uranium Enrichment and Stable Isotope Production
STATUS	TARGETABLE — TVEL SDN constructive coverage; Isotope JSC designation gap is the primary enforcement priority

KEY JUDGMENTS

KJ 1

ECP conducts uranium enrichment and stable isotope production at a single site. TVEL JSC's April 2024 SDN designation constructively covers ECP's enrichment operations. Whether ECP is explicitly listed requires OFAC confirmation.

LIKELY — HIGH CONFIDENCE

KJ 2

Isotope JSC — which distributes isotopes produced at ECP — is not OFAC-designated as of March 2026 and maintains active Western commercial relationships. Isotope JSC designation is the highest-priority unexecuted action at this node.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 3	Co-located enrichment and isotope separation at ECP creates material accountancy complexity for IAEA safeguards. The public record does not permit a firm conclusion on whether current safeguards mechanisms adequately address this dual-production configuration. ROUGHLY EVEN ODDS — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED

I. FACILITY OVERVIEW

ATTRIBUTION	The IAEA Country Nuclear Power Profiles and NTI Russia Profile identify the Electrochemical Plant (ECP) at Zelenogorsk — formerly Krasnoyarsk-45 — as a ZATO enrichment facility within the TVEL JSC enrichment network [TIER 1 — IAEA CNPP Russia 2024; TIER 2 — NTI 2023]. The facility produces enriched uranium and stable isotopes for industrial and medical applications.
JUDGMENT	Official and Tier 2 reporting supports a high-confidence judgment that ECP is an active enrichment node within the Russian fuel cycle. The isotope production dimension introduces an additional enforcement vector — through Isotope JSC — that does not exist at purely civilian enrichment sites.
LIMITS	The public record does not establish ECP's specific enrichment capacity as a share of the TVEL network, the specific isotopes produced and their declared end-use customers, or whether ECP has been explicitly listed under TVEL's SDN designation.

II. FUEL CYCLE CONNECTIONS

CONNECTION	DETAIL
UF_6 feedstock	From Seversk (SCC, Node 29) conversion or Angarsk (AECC, Node 31)
LEU enrichment output	Fuel fabrication at Elektrostal (Node 32) and Novosibirsk NCCP (Node 33)
Isotope output	Stable isotopes → domestic and international markets via Isotope JSC (not designated)
Geographic position	Trans-Siberian corridor between Seversk and Angarsk — eastern enrichment segment logistics hub

III. FINANCIAL NETWORK

ATTRIBUTION	ECP's enrichment revenue routes through TVEL JSC, designated by OFAC on April 19, 2024 [TIER 1 — OFAC SDN List]. Isotope JSC — a separate Rosatom commercial subsidiary — manages isotope sales from ECP and maintains Western commercial relationships with pharmaceutical and industrial customers. Isotope JSC is not designated as of March 2026.
JUDGMENT	Available reporting supports a high-confidence judgment that Isotope JSC's Western pharmaceutical and industrial customer relationships represent an active, undesignated financial flow from ECP operations through Western banking infrastructure. This is the primary enforcement gap at this node.
LIMITS	The public record does not establish the specific Western pharmaceutical companies purchasing isotopes from Isotope JSC's ECP-sourced inventory, the contract values, or the specific banking channels used for clearing.

IMPLEMENTATION FRICTION Isotope JSC operates under a separate commercial identity from TVEL. TVEL's SDN designation does not automatically capture Isotope JSC's financial flows. A separate Isotope JSC designation action or BIS Entity List action warrants attention to close this gap.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	ECP designatable as TVEL/Rosatom subsidiary. Explicit listing status unconfirmed.
CAATSA §231	Defense/Intel Sector	Rosatom sector designation covers ECP.
IEEPA / OFAC	Isotope JSC Designation	Isotope JSC not currently designated. Separate action required.
EAR / BIS	Entity List	Isotope JSC's U.S. commercial relationships create BIS Entity List exposure as alternative to OFAC designation.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC designated TVEL JSC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. Isotope JSC has not been designated. BIS maintains the Entity List under EAR authority; entities with proliferation-relevant commercial relationships are listable regardless of SDN status.
JUDGMENT	Official reporting supports a high-confidence judgment that Isotope JSC designation

	— under EO 13382 or as a BIS Entity List action — is immediately actionable, does not trigger the reactor safety constraint applicable to the enrichment supply chain, and would disrupt the most accessible Western financial flow from ECP operations.
LIMITS	The public record does not establish whether ECP is explicitly covered under TVEL's SDN designation or only constructively blocked. OFAC clarification warrants attention before concluding that the enrichment operations gap is fully closed.

VI. ENFORCEMENT GAPS

Gap 1 — Isotope JSC Not Designated

Isotope JSC operates an active Western commercial identity handling isotope sales from ECP and other Russian nuclear facilities. Its absence from the SDN list despite proliferation-relevant isotope sales is the most significant unaddressed enforcement gap at ECP.

Gap 2 — ECP Explicit SDN Listing Status Unclear

Whether ECP is explicitly listed under TVEL's SDN designation or only constructively covered requires OFAC confirmation. Explicit listing closes the legal ambiguity for U.S. persons and third-country compliance purposes.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	Isotope JSC Western pharmaceutical flows not traced from ECP specifically. EU pharmaceutical company regulatory disclosures may identify ECP-sourced isotope purchasing. Upgrade: European Medicines Agency regulatory filings for Russian-origin medical isotopes.
2	DESIGNATION	Explicit ECP SDN listing recommended. TVEL constructive coverage applies but Isotope JSC enrichment channel requires named-entity designation to close compliance ambiguity.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

ANGARSK — ANGARSK ELECTROCHEMICAL COMPANY

Conversion / Enrichment — IAEA International Uranium Enrichment Center Host

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

CONVERSION TIER

CONSTRAINED

NODE ID	31 — SENI-08
LOCATION	Angarsk, Irkutsk Oblast, Russia
COORDINATES	52.54°N, 103.89°E
OPERATOR	AECC / TVEL JSC / Rosatom
NETWORK GROUP	Russian Nuclear Fuel Cycle — Conversion and Enrichment Tier
PROCESS ROLE	Mid-Cycle — UF_6 Conversion and Enrichment; IAEA IUEC Host Site
STATUS	CONSTRAINED — IUEC multilateral architecture creates durable legal and diplomatic constraint on enforcement action

KEY JUDGMENTS

KJ 1

AECC hosts the IAEA International Uranium Enrichment Center (IUEC), established under a multilateral framework in 2007. Any enforcement action targeting AECC directly implicates the IUEC structure, triggering IAEA institutional involvement and potential UNSC escalation [TIER 1 — IAEA IUEC Agreement 2007].

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

The IUEC's IAEA association gives Russia a durable multilateral legitimacy argument against AECC designation that is structurally more difficult to overcome than the bilateral nuclear safety argument at Krasnokamensk.

VERY LIKELY — MODERATE-HIGH CONFIDENCE

KJ 3	A surgical OFAC designation carving out IUEC-associated activities from a broader AECC designation is legally feasible but operationally unusual and requires OFAC General Counsel coordination before execution. **ROUGHLY EVEN ODDS — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED**

I. FACILITY OVERVIEW

ATTRIBUTION	The IAEA's 2007 IUEC Agreement and subsequent IAEA Board of Governors documentation identify AECC as the host facility for the International Uranium Enrichment Center — a multilateral LEU assured-supply mechanism established to provide guaranteed enrichment services to states that forgo independent enrichment programs [TIER 1 — IAEA GOV/2006/14; IAEA CNPP Russia 2024].
JUDGMENT	Official IAEA reporting supports a high-confidence judgment that the IUEC's formal multilateral legal architecture at AECC creates a stronger and more durable enforcement constraint than exists at any other node in the Russian fuel cycle.
LIMITS	The public record does not establish the current IUEC membership, LEU delivery volumes under the IUEC framework, or the extent to which IUEC operations are financially separable from AECC's broader TVEL commercial enrichment activities.

DOCUMENTED STRUCTURAL CONSTRAINT The IUEC is a formally constituted IAEA-associated multilateral instrument with treaty-level documentation. Russia holds a permanent UNSC veto. Any enforcement action that disrupts IUEC operations would be characterized by Russia as an attack on a multilaterally-endorsed nonproliferation instrument, generating IAEA Board of Governors and UNSC escalation.

II. FUEL CYCLE CONNECTIONS

CONNECTION	DETAIL
U_3O_8 feedstock	Receives from Krasnokamensk (PIMCU, Node 27) and other ARMZ eastern mining operations
UF_6 conversion output	Internal enrichment cascades or transfer to other TVEL enrichment sites
LEU enrichment output	Fuel fabrication at Elektrostal (Node 32) and Novosibirsk NCCP (Node 33)
IUEC LEU supply	Assured supply to IUEC member states under declared multilateral framework
Trans-Siberian logistics	Eastern anchor of the Russian enrichment tier; connectivity to

	Krasnokamensk and Novosibirsk

III. FINANCIAL NETWORK

ATTRIBUTION	AECC's commercial enrichment revenue routes through TVEL JSC, designated by OFAC on April 19, 2024 [TIER 1 — OFAC SDN List]. The IUEC adds a separate financial layer: member state contributions and LEU sales revenue flow through a contractual structure partially subject to IAEA financial oversight. Rosatom retains operational and financial control over the IUEC.
JUDGMENT	Available reporting supports a moderate-confidence judgment that the IUEC financial architecture does not create genuine financial independence from Rosatom, but that enforcement actions disrupting IUEC financial flows would be characterized by Russia as violating the multilateral assured supply commitment.
LIMITS	The public record does not establish the specific financial separation between IUEC-associated and non-IUEC AECC commercial operations, or whether a surgical designation carving out IUEC-associated activities is achievable under OFAC licensing architecture.

IMPLEMENTATION FRICTION Financial enforcement actions targeting AECC/TVEL that disrupt IUEC operations will generate IAEA and member-state diplomatic pressure regardless of their legal validity. The enforcement leverage is real but operationally constrained by this multilateral response risk.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	AECC designatable as TVEL/Rosatom subsidiary. IUEC multilateral architecture creates political constraint.
CAATSA §231	Defense/Intel Sector	Rosatom sector authority covers AECC. Same political constraint applies.
IEEPA	Broad Economic Auth.	Available. IUEC complication most acute for broad IEEPA actions.
UNSC	UNSC Resolution	IUEC's IAEA association means enforcement could be elevated to UNSC level. Russia holds veto.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC designated TVEL JSC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. AECC is constructively covered as a TVEL subsidiary. The IUEC Agreement (IAEA GOV/2006/14) establishes the multilateral legal architecture that constrains enforcement action [TIER 1].
JUDGMENT	Official reporting supports a moderate-confidence judgment that AECC is legally designatable but that the IUEC multilateral constraint makes full unilateral designation operationally inadvisable without either a surgical carve-out or prior IUEC member state transition to alternative LEU sources.
LIMITS	The public record does not establish whether a surgical OFAC designation preserving IUEC-associated activities is achievable under current licensing architecture, or whether any IUEC member states have identified alternative LEU suppliers.

VI. ENFORCEMENT GAPS

Gap 1 — IUEC Multilateral Shield

The IUEC's IAEA association gives Russia a durable multilateral legitimacy argument against designation. Surgical designation carving out IUEC-associated activities is the only near-term workaround and requires unusual OFAC licensing architecture.

Gap 2 — IUEC Member State Transition Not Planned

No current mechanism exists for transitioning IUEC member states to alternative LEU suppliers as a precondition for AECC enforcement action. This gap should be addressed through allied nonproliferation diplomacy before enforcement sequencing reaches this node.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

IMPLEMENTATION FRICTION Financial pathway development: See WP-2026-SENI-FPA-01-v3 for traced pathways applicable to this node. Constructive SDN coverage applies through TVEL designation where applicable; explicit listing recommended before primary enforcement action.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

ELEKTROSTAL — METALLURGICAL PLANT

Nuclear Fuel Assembly Fabrication — VVER / RBMK / Research Reactors

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

CONVERSION TIER

TARGETABLE

NODE ID	32 — SENI-09
LOCATION	Elektrostal, Moscow Oblast, Russia
COORDINATES	55.79°N, 38.46°E
OPERATOR	JSC Metallurgical Plant Elektrostal / TVEL JSC / Rosatom
NETWORK GROUP	Russian Nuclear Fuel Cycle — Fabrication Tier
PROCESS ROLE	Downstream — Nuclear Fuel Assembly Fabrication (VVER, RBMK, Research Reactors)
STATUS	TARGETABLE — TVEL SDN constructive coverage; explicit listing and VVER-440/RBMK alternative fuel gap are primary constraints

KEY JUDGMENTS

KJ 1

Elektrostal is Russia's primary nuclear fuel assembly fabrication facility, the commercial delivery endpoint of the entire Russian nuclear fuel cycle. TVEL JSC's April 2024 SDN designation constructively covers Elektrostal. Explicit listing status requires OFAC confirmation.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Westinghouse has qualified alternative fuel for VVER-1000 designs at some Eastern European sites. No qualified alternative fuel supplier exists for VVER-440 and RBMK reactor designs as of March 2026. Enforcement actions affecting these fuel lines

	create reactor safety risk with no near-term resolution. **LIKELY — MODERATE-HIGH CONFIDENCE**
KJ 3	Reactor operator regulatory filings in EU, Finnish, and Asian jurisdictions provide partial visibility into active TVEL-Elektrostal fuel supply contracts — the most commercially transparent financial intelligence available in the Russian fuel cycle. **LIKELY — MODERATE CONFIDENCE**

I. FACILITY OVERVIEW

ATTRIBUTION	TVEL JSC public reporting and the World Nuclear Association's Fuel Market Report identify the Metallurgical Plant at Elektrostal as Russia's primary fuel assembly fabrication facility, producing VVER, RBMK, and research reactor fuel [TIER 1 — TVEL Annual Report; TIER 2 — WNA Fuel Report 2023]. The facility is located in Moscow Oblast — the only major fuel cycle facility in European Russia outside the Ural/Siberian ZATO complex.
JUDGMENT	Official and Tier 2 reporting supports a high-confidence judgment that Elektrostal is the commercial delivery endpoint of the Russian fuel cycle. Its Moscow Oblast location and commercial fuel assembly export function make it the most financially transparent node in the fabrication tier.
LIMITS	The public record does not establish the current complete export customer list for Elektrostal fuel assemblies post-2022, or whether any deliveries continue through third-country intermediaries to previously sanctioned customers.

II. FUEL CYCLE CONNECTIONS

CONNECTION	DETAIL
Enriched input	Receives enriched UF_6/UO_2 from UECC (Node 28), ECP (Node 30), SCC (Node 29), AECC (Node 31)
VVER export	Fuel assemblies → Russian-design VVER reactor operators globally (CNNC China, NPCIL India, Eastern European operators)
RBMK domestic	Fuel assemblies → Russian domestic RBMK reactors (Leningrad, Smolensk, Kursk NPPs)
Research reactor fuel	→ Rosatom international research reactor program
Alternative fuel competition	Westinghouse VVER-1000 alternative fuel qualified at some Eastern European sites as of 2024

III. FINANCIAL NETWORK

ATTRIBUTION	Elektrostal's financial flows are consolidated under TVEL JSC, designated by OFAC on April 19, 2024 [TIER 1 — OFAC SDN List]. TVEL's fuel assembly export contracts are the most commercially visible component of the Russian fuel cycle — contract values and counterparty identities are partially disclosed through reactor operator regulatory filings in jurisdictions with nuclear licensing transparency requirements [TIER 2 — Czech State Office for Nuclear Safety filings; Finnish TVO/Fortum disclosures].
JUDGMENT	Available reporting supports a moderate-confidence judgment that EU reactor operator regulatory filings provide actionable financial intelligence on active TVEL-Elektrostal contracts. This intelligence is available now and should be used to map clearing banks for each active contract.
LIMITS	The public record does not establish the full current export customer list for Elektrostal post-2022, or the extent to which deliveries continue through third-country intermediaries not visible in operator regulatory filings.

IMPLEMENTATION FRICTION VVER-440 and RBMK reactor types have no qualified alternative fuel supplier as of March 2026. Enforcement actions affecting these fuel lines create reactor safety risk for operators in Russia, Belarus, and remaining Eastern European VVER-440 sites with no near-term workaround.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	TVEL SDN April 2024. Elektrostal constructively covered; explicit listing recommended.
CAATSA §231	Defense/Intel Sector	Rosatom/TVEL sector designation covers downstream fabrication.
IEEPA / OFAC	Entity Designation	Explicit SDN listing would close ambiguity for U.S. persons and third-country compliance.
EAR / NRC	Nuclear Export Controls	NRC/DOE coordination required for actions affecting export fuel assemblies to operating reactors.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	OFAC designated TVEL JSC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. Westinghouse has qualified VVER-1000 alternative fuel at several Eastern European sites as of 2024 [TIER 2 — Westinghouse press releases; IAEA CNPP country reports]. No alternative fuel qualification exists for VVER-440 or RBMK designs.
JUDGMENT	Official reporting supports a moderate-high-confidence judgment that enforcement action at Elektrostal is viable for VVER-1000 customer contracts where Westinghouse alternative is qualified. Enforcement should be stratified by reactor type: prioritize VVER-1000 supply contracts first, address VVER-440 and RBMK supply lines only after alternative fuel qualification is achieved.
LIMITS	The public record does not establish the current timeline for VVER-440 or RBMK alternative fuel qualification, or whether any non-Russian supplier is actively pursuing qualification for these reactor types.

VI. ENFORCEMENT GAPS

Gap 1 — Explicit SDN Listing Status Unclear

Elektrostal is not confirmed as explicitly listed under TVEL's SDN designation. Constructive coverage under the 50% rule is legally sufficient but creates compliance ambiguity. Explicit listing requires OFAC confirmation or separate action.

Gap 2 — VVER-440 / RBMK Alternative Fuel Absent

No qualified alternative supplier exists for VVER-440 and RBMK fuel assemblies as of March 2026. Enforcement actions affecting these fuel lines create reactor safety risk for operators in Russia, Belarus, and remaining Eastern European VVER-440 sites. MODERATE-HIGH CONFIDENCE based on Tier 2 reporting from WNA and Westinghouse public disclosures.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	EU reactor operator filings confirm fuel assembly supply relationships; specific clearing banks not named. Upgrade: Czech SONS and Finnish STUK regulatory procurement disclosures for historical TVEL contracts.
2	DESIGNATION	Explicit Elektrostal SDN listing recommended for VVER-1000 supply enforcement. Constructive TVEL coverage applies; explicit listing eliminates ambiguity for EU operators with active contracts.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

NOVOSIBIRSK — CHEMICAL CONCENTRATES PLANT

Fuel Fabrication / Lithium-6 Production — Highest Proliferation Concern in Fabrication Tier

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

CONVERSION TIER

TARGETABLE

NODE ID	33 — SENI-10
LOCATION	Novosibirsk, Novosibirsk Oblast, Russia
COORDINATES	54.98°N, 82.90°E
OPERATOR	NCCP / TVEL JSC / Rosatom
NETWORK GROUP	Russian Nuclear Fuel Cycle — Fabrication Tier
PROCESS ROLE	Downstream — Nuclear Fuel Fabrication; Lithium-6 Production (Tritium Breeding Material)
STATUS	TARGETABLE — independent enforcement predicate via Li-6 production; Isotope JSC gap is immediately actionable

KEY JUDGMENTS

KJ 1

NCCP produces lithium-6 alongside civilian fuel fabrication. Lithium-6 is a tritium breeding material — when irradiated in a reactor it produces tritium, used as a thermonuclear booster in nuclear warheads — controlled under ECCN 1C233. This production provides an enforcement predicate entirely independent of the reactor fuel supply chain.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ

Isotope JSC distributes lithium-6 and other isotopes from NCCP and is not OFAC-

2	designated as of March 2026. Isotope JSC designation or BIS Entity List action is the highest-priority immediately actionable enforcement measure at this node. ALMOST CERTAINLY — HIGH CONFIDENCE
KJ 3	The available evidence is consistent with DPRK-adjacent networks having sought Russian lithium-6 through third-country intermediaries. The open-source picture remains incomplete; this assessment does not establish a documented procurement chain. LOW CONFIDENCE — ANALYTICALLY INFERRED. ROUGHLY EVEN ODDS — LOW CONFIDENCE — ANALYTICALLY INFERRED

I. FACILITY OVERVIEW

ATTRIBUTION	TVEL JSC annual reporting and the NTI Russia Nuclear Profile identify NCCP as Russia's second major nuclear fuel fabrication facility, producing VVER fuel assemblies and research reactor fuel [TIER 1 — TVEL Annual Report; TIER 2 — NTI 2023]. The facility produces lithium-6 isotope alongside fuel fabrication operations. Lithium-6 is controlled under BIS ECCN 1C233 as a nuclear-related material.
JUDGMENT	Official and Tier 2 reporting supports a high-confidence judgment that NCCP's lithium-6 production constitutes an independently actionable enforcement predicate that does not trigger the reactor safety arguments constraining action at other fuel cycle nodes.
LIMITS	The public record does not establish NCCP's annual lithium-6 production volume, the complete Isotope JSC customer list for lithium-6 sales, or end-use verification for declared civilian (battery technology, medical) applications.

II. FUEL CYCLE CONNECTIONS

CONNECTION	DETAIL
Enriched input	Receives enriched UF_6 from UECC (Node 28), ECP (Node 30), SCC (Node 29), AECC (Node 31)
VVER fuel output	Fuel assemblies → Asian and Siberian reactor operators; CNNC Chinese VVER fleet
Research reactor fuel	→ Rosatom international research reactor program
Li-6 output	→ Declared: industrial and battery markets via Isotope JSC (not designated)
Geographic logistics	Trans-Siberian rail → Russian Far East; land border connectivity to Kazakhstan/Mongolia

III. FINANCIAL NETWORK

ATTRIBUTION	NCCP's fuel fabrication revenue routes through TVEL JSC, designated by OFAC on April 19, 2024 [TIER 1 — OFAC SDN List]. Lithium-6 production revenue routes through Isotope JSC, a separate Rosatom subsidiary not currently designated. Isotope JSC maintains Western commercial relationships for medical and industrial isotope sales.
JUDGMENT	Available reporting supports a high-confidence judgment that Isotope JSC's lithium-6 sales channel represents an active, undesignated Western financial flow from NCCP operations. ECCN 1C233 export controls provide an immediately actionable BIS instrument that does not require OFAC SDN designation.
LIMITS	The public record does not establish the specific Western customers for Isotope JSC's NCCP-sourced lithium-6, the contract values, or the clearing banks used. The available reporting does not permit a firm conclusion on whether any lithium-6 sales have reached DPRK-adjacent purchasers.

IMPLEMENTATION FRICTION NCCP's location in commercial Novosibirsk — outside the ZATO regime — means it is more accessible to front-company procurement operations than the closed-city enrichment sites. End-use monitoring for NCCP's non-fuel commercial output is assessed as inadequate based on the absence of documented enforcement actions against identified purchasers.

IV. DESIGNATION AUTHORITIES

AUTHORITY	INSTRUMENT	NOTE
EO 13382	WMD Proliferators	NCCP designatable as TVEL/Rosatom subsidiary. Li-6 production provides independent WMD predicate.
CAATSA §231	Defense/Intel Sector	Rosatom sector designation covers NCCP.
IEEPA / OFAC	Isotope JSC Designation	Isotope JSC not currently designated. Separate designation action required.
EAR / BIS	ECCN 1C233 (Li-6)	Lithium-6 export controls. BIS Entity List and enhanced end-use monitoring immediately actionable.

V. ENFORCEMENT THEORY AND ACTION

ATTRIBUTION	Lithium-6 is listed as a controlled material under BIS ECCN 1C233 as a nuclear-related material. OFAC designated TVEL JSC on April 19, 2024, constructively covering NCCP [TIER 1 — OFAC SDN List; BIS CCL ECCN 1C233].
JUDGMENT	Official reporting supports a high-confidence judgment that NCCP presents the strongest independent proliferation predicate in the fabrication tier. The lithium-6 production capability is not required for civilian fuel fabrication and exists independently of the reactor fuel supply rationale, meaning enforcement action targeting NCCP's lithium-6 operations does not trigger the reactor safety arguments that constrain action at other nodes.
LIMITS	The public record does not establish a documented procurement chain connecting NCCP/Isotope JSC to DPRK-controlled entities. The DPRK nexus judgment is LOW CONFIDENCE — ANALYTICALLY INFERRED and cannot serve as a primary enforcement predicate without additional corroboration.

VI. ENFORCEMENT GAPS

Gap 1 — Isotope JSC Not Designated

Isotope JSC handles lithium-6 and other isotope sales from both NCCP and ECP. Its absence from the SDN list despite weapons-relevant isotope sales — specifically Li-6 under ECCN 1C233 — is the most significant unaddressed enforcement gap in the fabrication tier.

Gap 2 — DPRK Li-6 Procurement Evidence

The available evidence is consistent with DPRK-adjacent networks having sought Russian lithium-6, but the open-source picture remains incomplete. This assessment does not establish a documented procurement chain. LOW CONFIDENCE — ANALYTICALLY INFERRED. This assertion requires corroboration from Tier 1 or Tier 2 sources before use as a published enforcement predicate.

Gap 3 — End-Use Monitoring Inadequate

NCCP's location outside the ZATO regime makes it more accessible to front-company procurement than closed-city sites. End-use monitoring for NCCP's lithium-6 output is assessed as inadequate based on the absence of identified enforcement actions against verified purchasers.

VII. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

Evidentiary Development — Remaining Gaps

The following gaps define the distance between current OSINT-accessible evidence and execution-grade financial traceability. Each gap is linked to an upgrade path where applicable.

PRIORITY	GAP TYPE	DESCRIPTION AND UPGRADE PATH
1	FINANCIAL	Li-6 production financial flows not independently traced. FPA-01 v3 documents Isotope JSC Mo-99 pathway; Li-6-specific pathway is ANALYTICALLY INFERRED without named intermediaries. Upgrade: BIS export enforcement actions naming Li-6 purchasers;

		UN Panel DPRK proliferation annexes.
2	THRESHOLD	DPRK Li-6 procurement nexus is LOW CONFIDENCE — ANALYTICALLY INFERRED. This is the primary confidence constraint on NCCP designation. Upgrade: IC-collected DPRK procurement intelligence.
3	DESIGNATION	Isotope JSC designation would cover NCCP isotope operations through 50% rule. Explicit NCCP listing adds independent Li-6/ECCN 1C233 predicate. BIS Entity List action available unilaterally without OFAC coordination.

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